

Instructor Guide for *Making Simulations*

This will be the students' first lab, so they will need some hand holding. *Fortunately*, the goals for this session are pretty modest.

Key Concepts

To explore probability using Python this semester, the student will need a few basic skills. They will learn this foundation today:

- How to generate pseudorandom floating point numbers between 0 and 1.
- How to simulate flipping a coin a million times.
- How to simulate rolling a 100-sided die a million times.
- How to make a basic bar chart and histogram.

References

- The [standard random module](#) reference page
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Steps

0. The student needs to have a text editor and Python installed on their computer. If they don't know any text editors, VSCode is a common choice and runs on every platform. Make sure the Python is a recent version, at least 3.13.
1. In a text editor, the student will make a file called `random1.py`. They are trying to print 20 random numbers between 0 and 1. Here is a solution:

```
import random

for _ in range(20):
    print(random.random())
```

When it runs, it looks like this (but, of course, their numbers will be different):

```
$ python3 random1.py
```

```
0.4094994240162769
```

```
0.5866120149450935
```

```
...
```

```
0.16618582100435253
```

```
0.25114406877794404
```